

These lists include projects which release their software under open-source licenses and are related to artificial intelligence projects. These include software libraries, frameworks, platforms, and tools used for machine learning, deep learning, natural language processing, computer vision, reinforcement learning, artificial general intelligence, and more.

Agentic AI

Auto-GPT — experimental open-source autonomous GPT-based agent

Hermes Agent — self-improving AI agent developed by Nous Research

OpenClaw — autonomous task-executing AI agent

OpenHands — AI software development agent

Agentic AI frameworks

CrewAI — framework for building and orchestrating multi-agent AI systems

LangChain — framework for building applications and agents powered by large language models

Theia AI — framework for integrating AI capabilities and agents into integrated development environments (IDEs).

AI-assisted software development

Codex CLI – AI coding agent and command-line interface developed by OpenAI.

DeepSeek Coder – large language model designed for code generation and AI-assisted software development.

GLM – large language model family with capabilities for AI-assisted software development.

Grok Build – SpaceXAI's terminal-based AI coding agent.

OpenCode

Zed – source-code editor with integrated AI-assisted development and agentic coding

features.

AI hardware and inference acceleration

LLM inference and serving frameworks

llama.cpp — library that can perform inference on various LLMs such as Llama, Mistral, Gemma, DeepSeek or Qwen.

SGLang – framework for structured generation and high-performance LLM inference and serving

vLLM – high-throughput inference engine for large language models using techniques such as PagedAttention

Ollama – a software platform that enables easy deployment and interaction with many large language models locally.

Model formats and optimization toolkits

ONNX – Open Neural Network Exchange format for interoperability between AI frameworks

OpenVINO – Intel's toolkit for optimizing deep learning models for edge devices

TensorRT-LLM — Nvidia toolkit for optimizing and deploying large language models on GPUs

Artificial neural networks

EDLUT – event-driven neural network simulator for large-scale spiking networks

Emergent – cognitive modeling platform implementing neural networks

Encog – machine learning framework for Java and C# supporting neural networks

JOONE – Java-based neural network framework with modular architecture for learning tasks

Nengo – Python library for building and simulating large-scale neural systems

Neuroph – lightweight Java framework for creating neural networks

OpenNN – C++ library for designing, training, and deploying neural networks

SNNS – Stuttgart Neural Network Simulator, supports feedforward and recurrent neural networks

Convolutional neural networks (CNNs)

AlexNet — pioneering CNN for image classification, won the 2012 ImageNet competition

VGGNet — deep CNN known for its simplicity and use of 3x3 convolution filters

Inception — CNN architecture using parallel convolutional layers of different sizes

State-space models

Mamba – selective state-space model architecture for efficient sequence modeling

Chatbots

LAION OpenAssistant

Mycroft

Cognitive architectures and AGI platforms

ACT-R – cognitive architecture for modeling human cognition and developing cognitive agents.

CLARION – Connectionist Learning with Adaptive Rule Induction On-line, hybrid connectionist/symbolic cognitive architecture.

OpenCog – project that aims to build an open source artificial intelligence framework

Soar – cognitive architecture for decision-making and learning in intelligent agents

Computer vision and image processing

AForge.NET – computer vision, artificial intelligence, and robotics library for the .NET framework

Dlib – C++ library for computer vision and image processing

OpenCV — library of programming functions mainly for real-time computer vision

Point Cloud Library – library for point cloud processing

Tesseract – optical character recognition

Deep learning frameworks

BigDL – distributed deep learning library for Apache Spark

Caffe – deep learning framework focused on speed and modularity

Deeplearning4j – Java library for deep learning algorithms on the Java virtual machine

DeepSpeed – deep learning optimization library developed by Microsoft

fastai – deep learning library built on top of PyTorch

Fast Artificial Neural Network (FANN) – C library for feedforward neural networks

Horovod – distributed deep learning framework for TensorFlow, Keras, and PyTorch

Keras – Python library for artificial neural networks

Microsoft Cognitive Toolkit – deep learning framework developed by Microsoft Research

MXNet – deep learning framework for training and deploying deep neural networks

Neuroph – object-oriented artificial neural network framework written in Java

OpenNN – artificial neural network library written in C++

PlaidML – deep learning backend for neural networks and tensor computations

PyTorch – deep learning framework developed by Meta AI

PyTorch Lightning – high-level framework built on top of PyTorch for organizing and scaling deep learning models

TensorFlow – end-to-end open-source platform for machine learning and deep learning developed by Google Brain

Torch – scientific computing framework with support for machine learning and deep learning algorithms

Machine learning or data mining

Accord.NET — C# framework for machine learning, statistics, computer vision, and signal processing

Apache Mahout — scalable machine learning library for big data built on Hadoop and Spark

Apache SINGA — distributed deep learning and machine learning software

Apache SystemDS — ML system for the end-to-end data science lifecycle

CatBoost — machine learning library for gradient boosting on decision trees

Caffe — Image classification and image segmentation

Dlib — C++ machine learning and computer vision library

ELKI — data mining and unsupervised learning software

fastText — Word embeddings developed by Meta AI

Flux — machine learning library for the Julia programming language

Gensim — topic modelling and vector space model toolkit

JAX — machine learning library for numerical computing and automatic differentiation

H2O — machine learning and predictive analytics platform

Infer.NET — probabilistic programming framework for Bayesian inference

Jubatus — online machine learning and distributed computing framework

KNIME — modular data pipelining

LIBSVM — library for support vector machines

LightGBM — machine learning framework for gradient boosting

Massive Online Analysis — framework for machine learning and data mining on evolving data streams

Microsoft Cognitive Toolkit — deep learning framework developed by Microsoft Research

MindSpore — machine learning and deep learning framework

ML.NET — machine learning framework for .NET

mlpack — C++ header-only machine learning library

MXNet — deep learning framework

Orange — machine learning, data mining, data visualization, and data analysis

PyMC — Python framework for Bayesian modeling and probabilistic programming

Scikit-learn — library built on top of NumPy, SciPy, and matplotlib

Scikit-multiflow — Python framework for online and stream machine learning

Shogun — C++ library for large-scale machine learning

Spark MLlib — machine learning library for Apache Spark

TensorFlow — machine learning and deep learning platform

Theano — numerical computation library for machine learning

Vowpal Wabbit — machine learning system for fast online learning

Weka — collection of machine learning algorithms for data mining tasks

XGBoost — machine learning library for gradient boosting

AutoML platforms

TPOT – tree-based pipeline optimization tool using genetic programming

Neural Network Intelligence – Microsoft toolkit for hyperparameter tuning and neural architecture search

MindsDB – AutoML platform that embeds machine learning into SQL databases and applications

Language models

Apertus – Swiss National AI Initiative LLM

BERT – Google LLM

BLOOM – BigScience multilingual LLM

Cerebras-GPT – Cerebras Systems LLMs

DeepSeek – R1 and V3 models

Gemma 4 – Google LLM

GLM-4.5 and later versions – Z.ai LLMs

GPT-1, GPT-2, and GPT-OSS – OpenAI LLMs

GPT-J, GPT-Neo, and GPT-NeoX – EleutherAI LLMs

Granite – IBM LLMs

Grok-1 – xAI LLM

Inkling – Thinking Machines

Kimi (some versions) – Moonshot AI LLMs

Mistral (some versions)

Muse Glimmer — Meta Superintelligence Labs LLM

Nemotron 3 – Nvidia open-weight models

OLMo – Allen Institute for AI LLM

Phi – Microsoft LLMs

Qwen (some versions) – Alibaba Cloud LLMs

Sarvam-M, Sarvam-105B and Sarvam-30B – Sarvam AI LLMs

SmolLM – Hugging Face small language models (SLMs)

Step-3.5-Flash – StepFun LLM

T5 – Google LLM

XLNet – Google LLM

Natural language processing (NLP)

Apache OpenNLP

Apertium – rule-based machine translation platform.

ChatScript – natural language engine and dialog management system

General Architecture for Text Engineering – information extraction

Gensim – topic modeling and document similarity analysis library

GloVe – unsupervised learning algorithm for obtaining vector representations of words

Mallet – Java "Machine Learning for Language Toolkit"

MontyLingua – libraries and programs for symbolic and statistical NLP for both Python and Java

Moses – statistical machine translation engine to train statistical models of text from a source language to a target language

NiuTrans – statistical machine translation

NLTK – natural Language toolkit for symbolic and statistical NLP

Probabilistic Action Cores – interpreter for natural-language instructions for robotic applications

spaCy – Python library

Spark NLP – text processing library for advanced NLP for Python, Java, and Scala.

Word2vec – obtaining vector representations of words

Speech recognition systems

CMU Sphinx

DeepSpeech

Julius

Whisper

Reactive planning

GOLOG – logic programming language, situation calculus, first-order logical language for reasoning about action and change.

Reinforcement learning frameworks

KataGo – reinforcement learning agent designed for playing the game of Go

Robotics software

ArduPilot

CoppeliaSim

Gazebo

Mobile Robot Programming Toolkit

OpenRTM-aist

Paparazzi Project

Player Project

Python Robotics

Robot Operating System

TurtleBot

Webots

Text to image

Flux

Stable Diffusion

Text to speech

Festival Speech Synthesis System

WaveNet

eSpeak

Transformer libraries

Hugging Face transformers library – Python library of pretrained transformer models for NLP,

computer vision, speech, and more.

AI-assisted algorithm discovery

AlphaTensor – DeepMind reinforcement-learning system for discovering efficient matrix multiplication algorithms

Music and audio generation

Riffusion – generating music and audio using Stable Diffusion.

Knowledge bases and knowledge graphs

ConceptNet – multilingual commonsense knowledge graph used in natural-language processing and artificial-intelligence applications.

DBpedia – project that extracts structured knowledge from Wikipedia using an open-source extraction framework.

See also

Open-source artificial intelligence

List of artificial intelligence algorithms

List of artificial intelligence journals and List of artificial intelligence books

List of artificial intelligence projects

List of free and open-source software packages for artificial intelligence

List of proprietary artificial intelligence software

Open weights

Common Crawl – nonprofit that crawls the web and freely provides its archives and datasets to the public under an MIT License

Google Colab – an O-IDE Jupyter notebook environment with free access to GPUs and TPUs for

machine learning and deep learning development

Applications of artificial intelligence

References

External links

opensource.org/ai – Open Source Initiative

Data and AI White paper (July 24, 2024) – Open Data Institute